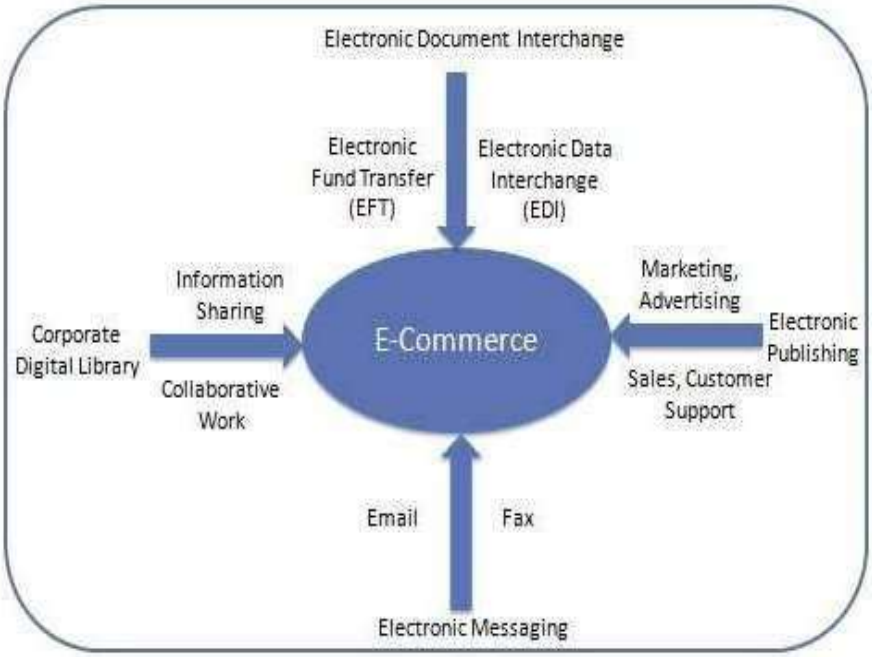
Unit 1: Introduction to Electronic-Commerce (E-commerce)

➲ 1.1 E-Commerce (EC) overview

- ✓ **Definition:** E-commerce means buying and selling products or services using the Internet or other computer networks.
- ✓ **Purpose:** E-commerce helps businesses, sellers, and customers save money, get better quality products and services, and get things faster.
- ✓ **Paperless Information Sharing:** E-commerce uses different ways to share business information without paper, like:

  - **EDI (Electronic Data Exchange)**
  - **Email**
  - **Online Bulletin Boards**
  - **EFT (Electronic Fund Transfer)**
  - **Other Network-based Methods**



- ✓ **EDI (Electronic Data Exchange)**: A way for businesses to send documents like orders or invoices electronically.

  Example: A store sends a purchase order to a supplier using EDI instead of a paper form.

- ✓ **Email**: Sending and receiving business information through electronic mail.

  Example: A company sends an invoice or update to a customer via email.

- ✓ **Online Bulletin Boards**: Websites where businesses post important information or updates.

  Example: A company shares news about a new product launch on its online bulletin board.

- ✓ **EFT (Electronic Fund Transfer)**: Transferring money between bank accounts electronically.

  Example: A customer pays for an online purchase using a bank transfer or digital wallet.

- ✓ **Other Network-based Methods**: Using various online tools and technologies to share business information.

  Example: Using chat apps or specialized business platforms to send and receive documents or updates.

➲ 1.2 Features/Aims of E-commerce

E-Commerce provides following features:

- ✓ **Non-Cash Payment**: E-commerce allows payments using credit/debit cards, smart cards, and online bank transfers.

  Example: Buying clothes online and paying with a credit card.

- ✓ **24x7 Service Availability**: Businesses can serve customers anytime, any day.

  Example: Shopping online at midnight or on a holiday.

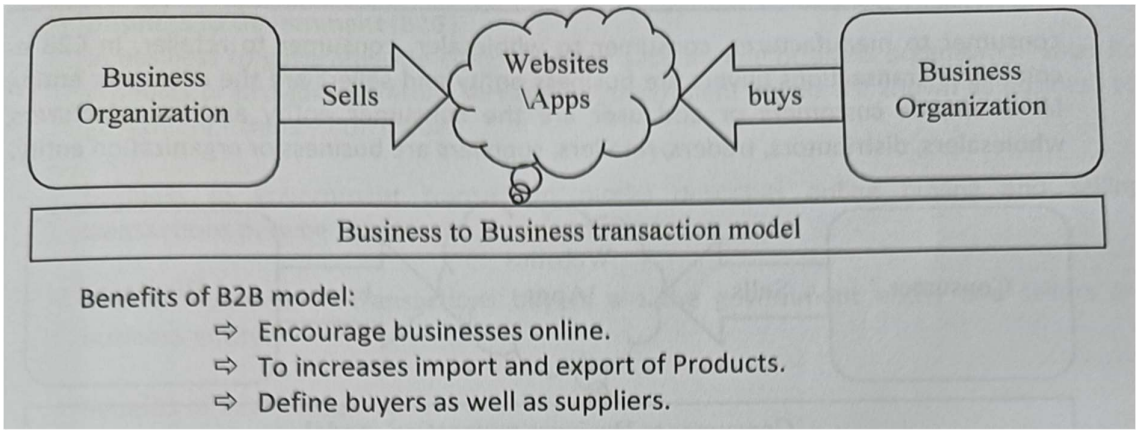
- ✓ **Advertising/Marketing**: E-commerce helps businesses reach more customers and market products better.
  Example: Online ads on social media or websites.
- ✓ **Improved Sales**: Orders can be placed anytime without human help, leading to more sales.
  Example: Ordering food online at any hour.
- ✓ **Support**: E-commerce offers pre-sales and post-sales support for better customer service.
  Example: Chat support for questions before buying and help after purchase.
- ✓ **Inventory Management**: E-commerce automates product inventory and generates reports easily.
  Example: Checking product stock online instantly.
- ✓ **Communication Improvement**: E-commerce enables faster and more reliable communication with customers and partners.
  Example: Quick updates via email or messaging apps.

❖ Types of e-commerce

1) **Business-to-Business [B2B]**
2) **Business-to-Consumer [B2C]**
3) **Consumer-to-Business [C2B]**
4) **Consumer-to-Consumer [C2C]**

❖ Business-to-Business [B2B]

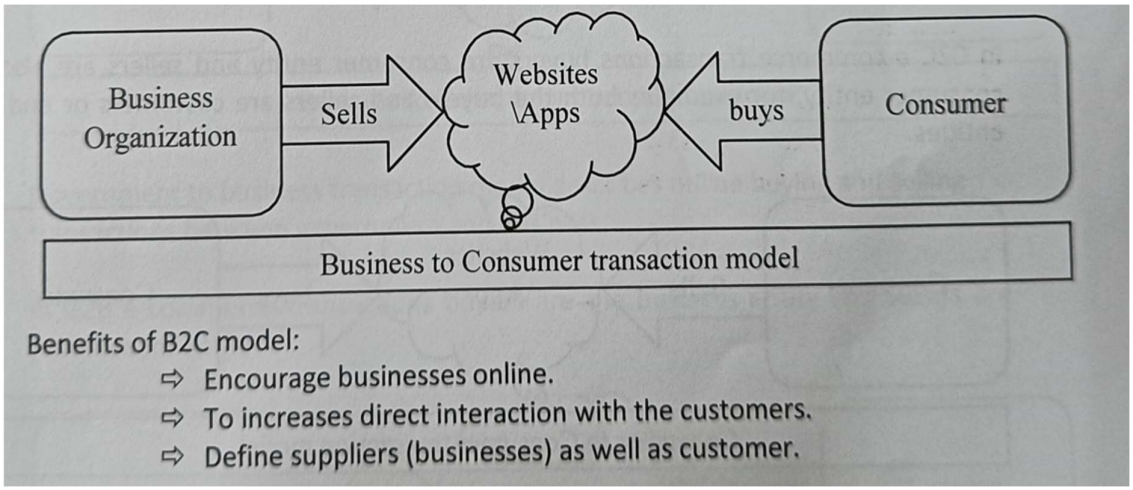
- ✓ **Definition**: B2B (Business-to-Business) is when one business sells products or services to another business.
- ✓ **Types of Deals**: This can include products, goods, or services.
- ✓ **Example**: Google collaborating with LG to create a mobile phone.



❖ **Business-to-Consumer [B2C]**

✓ **Definition**: B2C (Business-to-Consumer) is when a business sells products or services directly to individual consumers or customers.

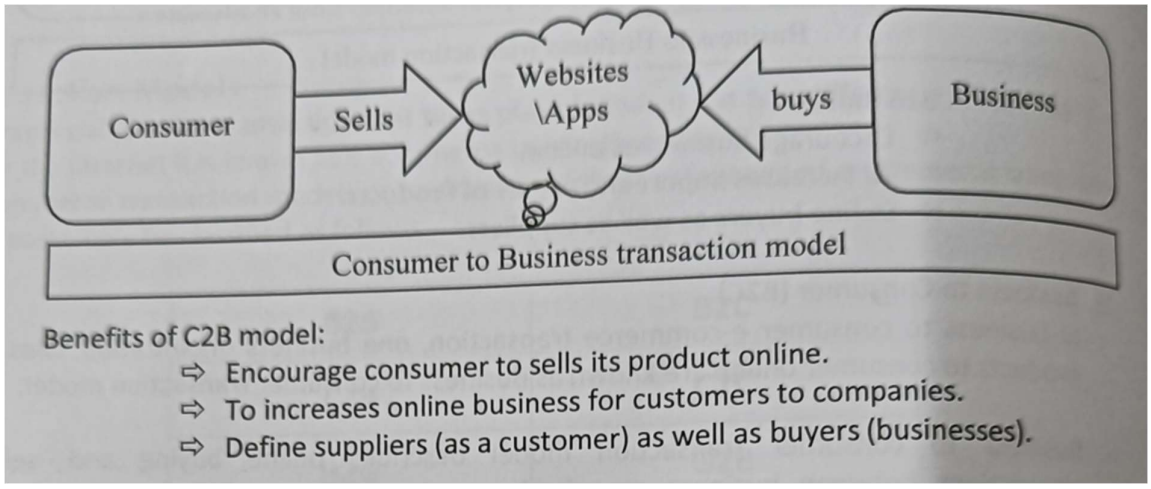
✓ **Example**: Shopping on websites like flipkart.com, amazon.com, or using services from www.icicibank.com.



❖ **Consumer-to-Business [C2B]**

✓ **Definition**: C2B (Consumer-to-Business) is when a customer offers services or works for a business.

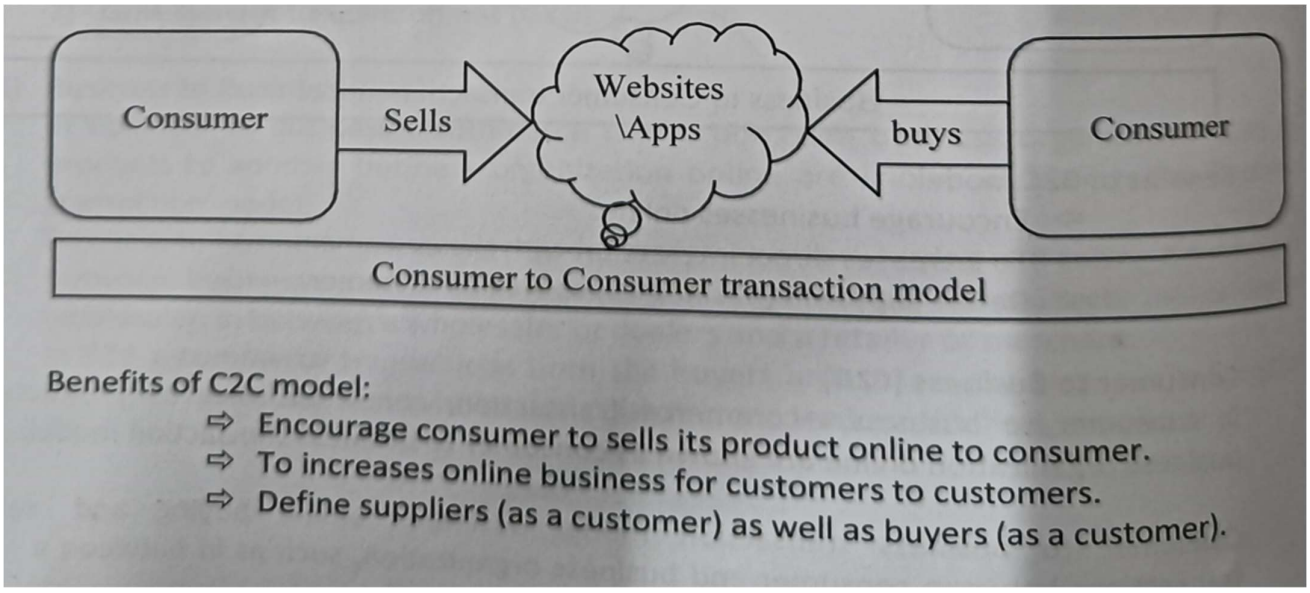
✓ **Example**: Websites like priceline.com, where customers can offer their services or set prices for products and services.



❖ **Consumer-to-Consumer [C2C]**

✓ **Definition**: C2C (Consumer-to-Consumer) is when a consumer sells products or services directly to another consumer.

✓ **Example**: Websites like olx.com and quicker.com, where people can buy and sell items to each other.



➲ Advantages and Limitations of E-Commerce

- E-commerce benefits to E-Commerce advantages can be broadly classified in three major categories:
  - Advantages to Organizations
  - Advantages to Consumers
  - Advantages to Society

❖ **Benefits of E-Commerce for Organizations:**

- ✓ **Global Reach**: Expands business to national and international markets with low investment.
- ✓ **Cost Savings**: Reduces costs by digitizing information and processes.
- ✓ **Better Service**: Allows better interaction and service for more customers.
- ✓ **Improved Supply Chain**: Increases productivity and supports on-demand production.
- ✓ **24/7 Availability**: Business can run all day, every day, without extra costs.
- ✓ **Quick Implementation**: Faster launch of new ideas due to better customer communication.
- ✓ **Better Inventory Management**: Reduces costs related to managing stock.

❖ **Benefits of E-Commerce for Customers:**

- ✓ **24x7 Support**: Customers can buy products or ask about services anytime and from anywhere.
- ✓ **More Choices**: E-commerce gives more options and faster delivery.
- ✓ **Lower Prices**: Customers can compare prices to find cheaper deals.
- ✓ **Product Reviews**: Customers can read reviews from others before buying.
- ✓ **Quick Information**: Customers can get detailed product info right away.
- ✓ **No Sales Tax**: In many places, buying online means no sales tax.

❖ Benefits of E-Commerce for Society:

- ✓ **Telecommuting**: People do not need to travel to shop, which reduces traffic and air pollution.
- ✓ **Help for the Poor**: E-commerce lowers product costs, making it easier for people in rural areas to buy things.
- ✓ **Better Living Standards**: Online shopping from home can improve people's quality of life.
- ✓ **Public Services**: E-commerce helps the government provide services like health care, education, and social support more efficiently and at lower costs.

❖ Disadvantages/Limitations of E-Commerce:

1. Technical Disadvantages:

- ✓ **Security Issues**: E-commerce can have problems with safety and reliability because it might not be set up well.
- ✓ **Fast Software Changes**: Software keeps changing quickly, which can be hard to keep up with.
- ✓ **Limited Internet Speed**: In some countries, the internet speed is slow or not strong enough.

2. Non-Technical Disadvantages:

- ✓ **High Initial Cost**: Building an e-commerce site can be very expensive and may take longer if there are mistakes or lack of experience.
- ✓ **Trust Issues**: Customers may not trust unknown sellers and may prefer shopping in physical stores.
- ✓ **Product Quality**: Online shopping does not let you touch or feel the products before buying.

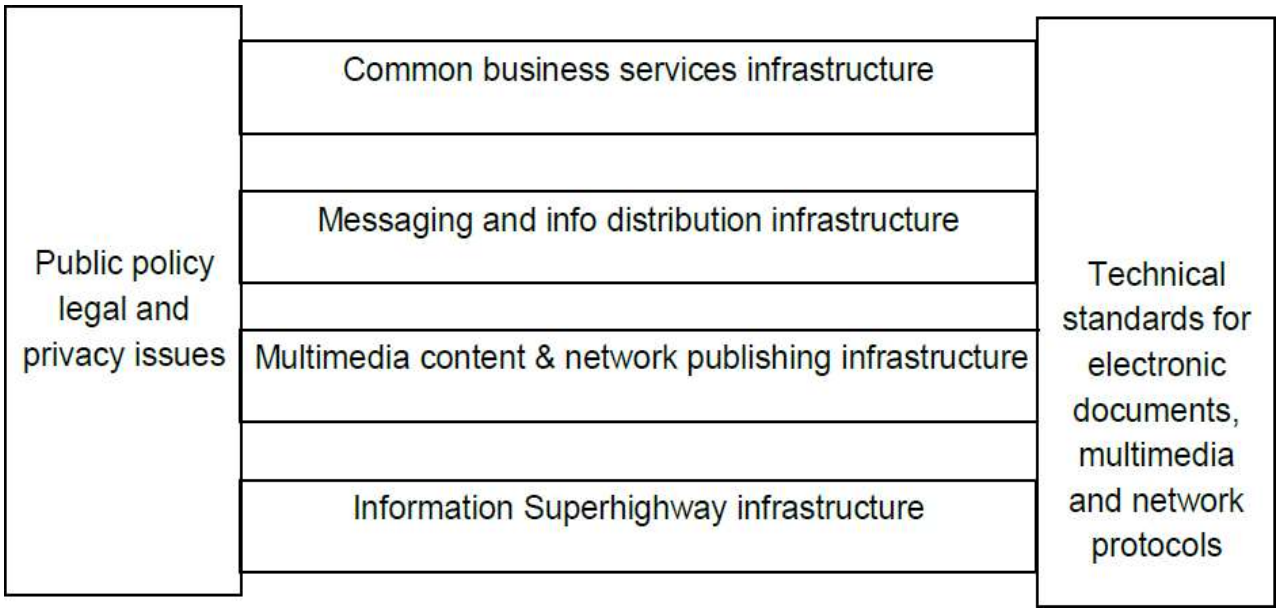
- ✓ **Internet Access**: The internet is not cheap or easy to use for some people, especially in remote areas.

➲ 1.3 Framework of E-Commerce

- **Base**: E-commerce runs on computers, networks, and software, creating an "information highway" (i-way).

- **Two Main Pillars**:

1. **Applications**: Tools and software for e-commerce.
2. **Infrastructure**: Systems and technology supporting these tools.



❖ **<u>Framework of E-Commerce:</u>**

1. **Public Policy and Privacy Issues:**

✓ Sets rules for businesses and customers in e-commerce.

2. **Technical Standards:**

✓ Defines formats for information (like multimedia) and network communication rules.

➢ **Four Main Infrastructures in E-Commerce:**

1. **Business Services Infrastructure:**
   - Handles buying, selling, security, and online payments.
2. **Messaging Infrastructure:**
   - Helps in sending and receiving information.
3. **Multimedia and Publishing Infrastructure:**
   - Creates content using audio, video, and images.
4. **Information Highway (I-way):**
   - Moves e-commerce data through the internet, telecom, and cable.
   - Needed for business and customer communication.

✸ **Examples:**

- **Digital Games:** Combines music, video, and software.
- **E-books:** Includes text, pictures, and music.

✸ **How It Works:**

- Just like trucks transport goods, I-way uses email, or file transfers to send data.

➲ 1.4 Electronic Commerce <u>Consumer</u> Applications

E-commerce offers various applications for consumers, including:

✓ **Entertainment:**

- Movies on demand, online video catalogs, interactive ads, multiplayer games, and online communities.

✓ **Financial Services:**

- Online banking, investment services, and financial news updates.

✓ **Essential Services:**

- Online shopping, e-catalogs, telemedicine (remote medical help), and health checkups.

➲ 1.5 Electronic Commerce Organization Applications:

E-commerce helps businesses in these ways:

1. **Understanding Customers:**
   - Study customer behavior to attract new customers and keep existing ones.
2. **Delivering Products:**
   - Focus on packaging, shipping, and online platforms to send products easily.
3. **Adapting to Changes:**
   - Be flexible to switch partners, platforms, or networks anytime.
4. **Marketing:**
   - **Target Marketing:** Focus on specific customer groups.
   - **Relationship Marketing:** Build strong, long-term customer bonds.
   - **Virtual Inventory:** No need to keep stock, sell everything online.

♣ E-Commerce - Business Models

▪ E-Commerce or Electronics Commerce business models can generally categorized in following categories.

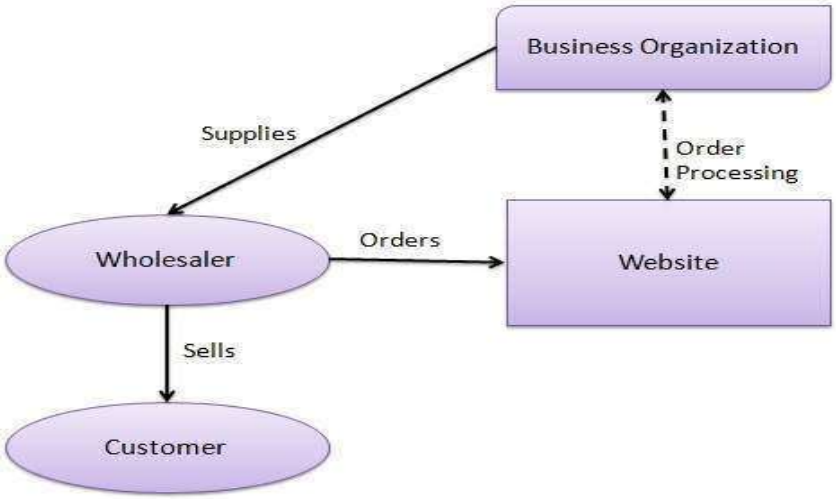
- **B**usiness - to - **B**usiness (B2B)
- **B**usiness - to - **C**onsumer (B2C)
- **C**onsumer - to - **C**onsumer (C2C)
- **C**onsumer - to - **B**usiness (C2B)

❖ Business - to - Business (B2B)

✓ **Definition:** B2B is when one business sells products or services to another business.

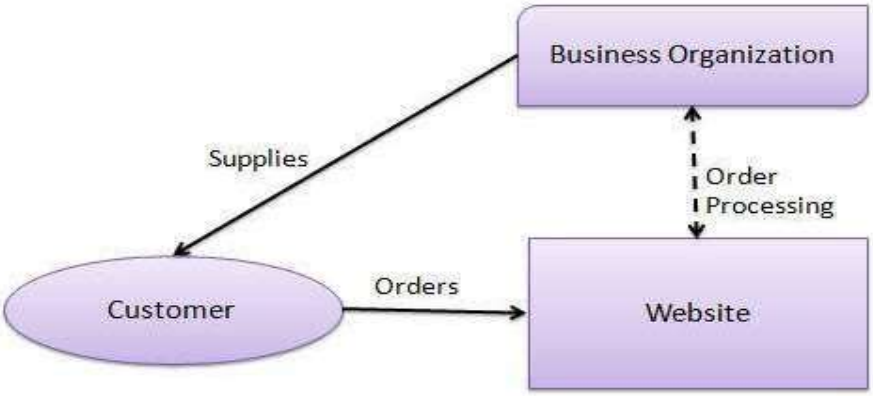
- ✓ **How it Works:** A B2B website sells products to an intermediary, who then sells them to the final customer.

- ✓ **Examples of B2B Businesses:** Software companies, office furniture suppliers, and document hosting services.



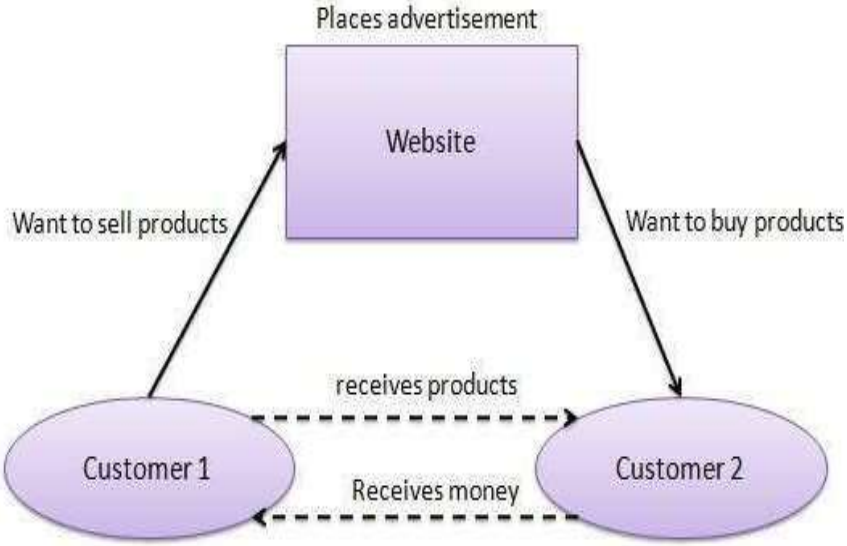
❖ **Business - to - Consumer (B2C)**

- ✓ **Definition:** A website sells products directly to customers.
- ✓ **How it Works:** Customers browse the website, pick a product, and place an order.
- ✓ **Order Process:** The website sends a notification to the business, and the business ships the product to the customer.
- ✓ Examples of B2C businesses are everywhere.



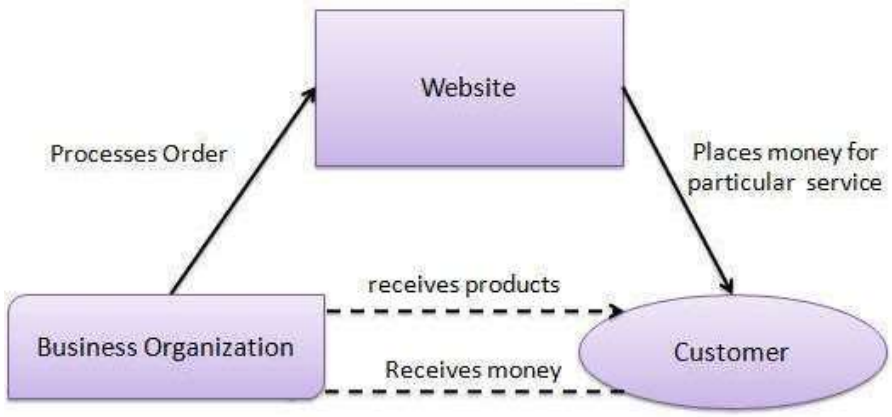
❖ Consumer - to - Consumer (C2C)

- ✓ **Definition:** A website allows consumers to sell or rent their items, like homes, cars, or rooms, by posting information online.
- ✓ **Cost:** The website may charge a fee or provide the service free.
- ✓ **How it Works:** Another consumer can see the listing and decide to buy the item.
- ✓ **Growth and Planning:** C2C websites became popular with the rise of e-commerce and increased trust in online shopping. They help consumers' trade items and earn a small fee for the site owner. Starting a C2C site needs careful planning.



❖ Consumer - to - Business (C2B)

- ✓ **Definition:** In this model, consumers use a website to compare or offer goods and services to businesses.
- ✓ **How it Works:** Consumers specify how much they are willing to spend for a service or product. For example, checking and comparing interest rates for personal or car loans from different banks.
- ✓ **Example:** Google AdSense is an example of C2B, where consumers provide content or services that businesses use for advertising.



➲ 1.6 Mobile Commerce (M-Commerce):

- ✓ **Definition:** M-commerce refers to transactions done through mobile devices such as smartphones, tablets, or wearables.
- ✓ **Purpose:** It allows consumers to buy and sell goods and services online anytime and anywhere.
- ✓ **Examples of M-Commerce Innovations:**

  - Booking tickets online
  - Mobile banking and money transfers
  - Contactless payments and in-app purchases
  - Buying digital content
  - Location-based services
  - Mobile marketing with coupons and loyalty cards

- ✓ **Types of Mobile Commerce Applications:**

  - **Mobile Shopping:** Similar to e-commerce but designed for mobile use. This includes dedicated apps, mobile-optimized websites, and social media platforms like Facebook or Instagram for in-app shopping.

- **Mobile Banking:** Similar to online banking but may have some limitations in transactions on mobile devices. Often involves dedicated apps, although some financial companies use chatbots or messaging apps for customer service.
- **Mobile Payments:** Focuses on innovations in payment methods using mobile devices. Includes options beyond mobile wallets, offering various ways for consumers to make payments.